

Diffracted radiance: a fundamental quantity in nonparaxial scalar diffraction theory

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Most authors include a paraxial (small-angle) limitation in their discussion of diffracted wave fields. This paraxial limitation severely limits the conditions under which diffraction behavior is adequately described. A linear systems approach to modeling nonparaxial scalar diffraction theory is developed by normalization of the spatial variables by the wavelength of light and by recognition that the reciprocal variables in Fourier transform space are the direction cosines of the propagation vectors of the resulting angular spectrum of plane waves. It is then shown that wide-angle scalar diffraction phenomena are shift invariant with respect to changes in the incident angle *only* in direction cosine space. Furthermore, it is the diffracted *radiance* (not the intensity or the irradiance) that is shift invariant in direction cosine space. This realization greatly extends the range of parameters over which simple Fourier techniques can be used to make accurate calculations concerning wide-angle diffraction phenomena. Diffraction-grating behavior and surface-scattering effects are two diffraction phenomena that are not limited to the paraxial region and benefit greatly from this new development. © 1999 Optical Society of America

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1. Introduction

It is well known that the irradiance distribution on a plane in the far field (the Fraunhofer region) of a diffracting aperture is given by the squared modulus of the Fourier transform of the complex amplitude distribution emerging from the diffracting aperture^{1,2}:

$$E(x_2, y_2) = \frac{1}{\lambda^2 z^2} \left| \mathcal{F}\{U_0^+(x_1, y_1)\} \right|_{\xi=\frac{x_2}{\lambda z}, \eta=\frac{y_2}{\lambda z}}^2, \quad (1)$$

where $U_0^+(x_1, y_1) = U_0^-(x_1, y_1)t_1(x_1, y_1)$ is the complex amplitude distribution emerging from the diffracting aperture with complex amplitude transmittance $t_1(x_1, y_1)$ and $\mathcal{F}\{\cdot\}$ denotes the Fourier transform operation

$$\mathcal{F}\{U_0^+(x_1, y_1)\} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} U_0^+(x_1, y_1) \exp[-i2\pi(x_1\xi + y_1\eta)] dx_1 dy_1. \quad (2)$$

The spatial frequencies ξ and η are the reciprocal variables in Fourier transform space. Also, the Fresnel diffraction integral is given by the Fourier transform of the product of the aperture function with a quadratic phase factor.^{1,2} Implicit in both the Fresnel and the Fraunhofer approximations is a *paraxial* limitation that restricts their use to small diffraction angles and small angles of incidence.² This paraxial limitation severely restricts the conditions under which diffraction phenomena are adequately described.

In this paper a linear systems approach to modeling nonparaxial diffraction behavior for arbitrary obliquely incident beams is developed. It is shown that the diffracted *radiance* is the fundamental quantity that exhibits linear, shift-invariant behavior with respect to the incident angle when formulated in terms of the direction cosines of the propagation vectors. Because diffraction gratings are inherently wide-angle devices that are frequently used with large oblique incident angles and scattering effects from rough surfaces are another diffraction phenomenon that is not limited to the paraxial region, these two practical examples are discussed in detail. These examples show that many of the limitations usually attributed to scalar diffraction theory are really the result of the unnecessary paraxial approximation and not a limitation inherent to scalar theory.

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2. Historical Background

Harvey³ generalized Goodman's Fourier treatment of the scalar diffraction theory to include new insight into the phenomenon of diffraction throughout the whole space in which it occurs. By use of a scaled coordinate system in which all the spatial variables are normalized by the wavelength of the light,

$$\hat{x} = x/\lambda, \quad \hat{y} = y/\lambda, \quad \hat{z} = z/\lambda, \quad \text{etc.}, \quad (3)$$

the reciprocal variables in Fourier transform space are precisely the direction cosines α and β of the propagation vectors of the plane-wave components

$$\alpha = \hat{x}/\hat{r}, \quad \beta = \hat{y}/\hat{r}, \quad \gamma = \hat{z}/\hat{r}, \quad (4)$$

where

$$\hat{r} = (\hat{x}^2 + \hat{y}^2)^{1/2}$$

in the angular spectrum of plane waves discussed by Ratcliff,⁴ Goodman,² and Gaskill.¹ Recall that this angular-spectrum approach leads to a transfer function of free space whose Fourier transform (the impulse response of the diffraction process) is a precise mathematical description of a Huygens wavelet, complete with cosine obliquity factor and a $\pi/2$ phase delay.^{2,3} The convolution of that impulse response with the optical disturbance emerging from a diffracting aperture is precisely the familiar Rayleigh–Sommerfeld diffraction integral for near-field ($z \gg \lambda$) diffraction:

$$U(\hat{x}_2, \hat{y}_2; \hat{z}) = -i \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} U_0(\hat{x}_1, \hat{y}_1; 0) \times \frac{\hat{z} \exp(i2\pi\hat{l})}{\hat{l}} d\hat{x}_1 d\hat{y}_1. \quad (5)$$

The Rayleigh–Sommerfeld diffraction integral is rather unwieldy to solve explicitly for most problems of practical interest. The Fresnel and the Fraunhofer diffraction formulas are obtained by the retention of only the first two terms in the binomial expansion for the quantity $\hat{l}$ in the exponent of the Rayleigh–Sommerfeld diffraction integral. These explicit approximations impose severe restrictions on both the size of the aperture (relative to the observation distance) and the diffraction angle. In fact, Goodman² states that the Fresnel approximation is equivalent to a paraxial approximation. So that we do not impose these restrictions, *all* terms from the binomial expansion must be retained. This can be accomplished by means of rewriting Eq. (5) as a Fourier transform integral of a generalized pupil function that includes phase variations that resemble conventional aberrations.⁵ Any departures of the actual diffracted wave field from that predicted by the Fourier transform of the aperture function are shown to have the same functional form as the conventional wave-front aberrations of imaging systems. These aberrations, which are inherent to the diffraction process, are precisely the effects that are ignored when

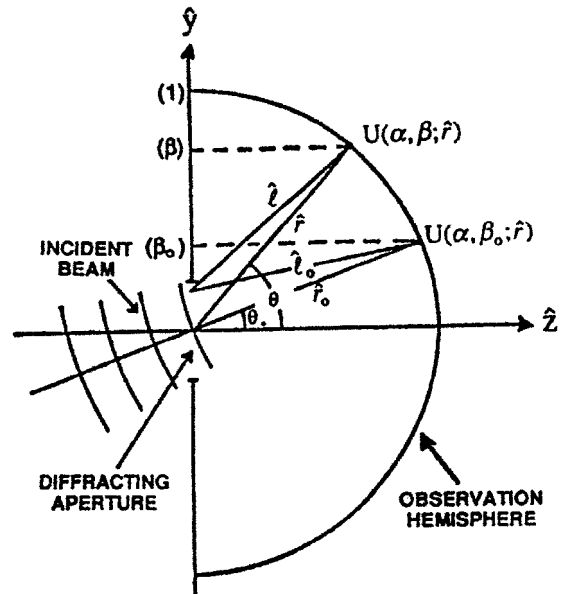


Fig. 1. Geometrical configuration of the incident beam, the diffracting aperture, and the observation hemisphere.

making the usual Fresnel and Fraunhofer approximations. Significant insight is obtained by recognition that the impulse response of the diffraction process is merely a Huygens wavelet and that near-field diffraction patterns are merely *aberrated* Fraunhofer diffraction patterns.

When a spherical wave is incident upon a diffracting aperture and the observation space is a hemisphere centered on the aperture the phase variations mentioned above are frequently negligible.³ The diffracted wave field on the hemisphere is then given directly by the Fourier transform of the aperture function:

$$U(\alpha, \beta; \hat{r}) = \gamma[\exp(i2\pi\hat{r})/(i\hat{r})]\mathcal{F}\{U_0(\hat{x}, \hat{y}; 0)\}. \quad (6)$$

Furthermore, this Fourier transform relation is valid not merely over a small region about the optical axis but also over the *entire* hemisphere (with certain restrictions, depending on the residual phase variations).⁵

Now consider the situation in which the incident radiation strikes the diffracting aperture at an angle θ_0 , as illustrated in Fig. 1. This is equivalent to introducing a linear phase variation across the aperture and attenuating the irradiance in the plane of the aperture by a factor of $\gamma_0 = \cos(\theta_0)$. By applying the shift theorem of Fourier transform theory to Eq. (6), we find that the complex amplitude distribution in direction cosine space is a function of $\beta - \beta_0$:

$$U(\alpha, \beta - \beta_0; \hat{r}) = \gamma[\exp(i2\pi\hat{r})/(i\hat{r})]\mathcal{F}\{U_0'(\hat{x}, \hat{y}; 0) \times \exp(i2\pi\beta_0\hat{y})\}, \quad (7)$$

where

$$U_0'(\hat{x}, \hat{y}; 0) = \sqrt{\gamma_0}U_0(\hat{x}, \hat{y}; 0).$$

Here β is the direction cosine of the position vector of the observation point and β_0 is the direction cosine of the position vector of the undiffracted beam. Note that the direction cosines are obtained by the mere projection of the respective points on the hemisphere back onto the plane of the aperture and normalization to a unit radius. The complex amplitude distribution at an arbitrary point on the hemisphere can now be said to be a function of the distance of the observation point from the undiffracted beam in direction cosine space. Furthermore, $\gamma = \cos \theta$ is just a cosine obliquity factor. Hence nonparaxial diffraction phenomena are shown to be linear, shift-invariant phenomena with respect to the incident angle if we formulate the problem in terms of the direction cosines of the propagation vectors.

3. Diffracted Radiance: The Fundamental Quantity Predicted by Scalar Theory

Before proceeding to show that radiance is the fundamental quantity that is shift invariant with respect to the incident angle in a nonparaxial scalar diffraction theory, we briefly review the definitions of a few radiometric quantities. In the past, scientists generally have used the word intensity to mean the flow of energy per unit area per unit time. However, by international, if not universal, agreement that term is slowly being replaced by the word irradiance⁶:

$$\text{Irradiance} \equiv E = \frac{\partial P}{\partial A_c}, \quad (\text{watts/area}). \quad (8)$$

Irradiance is thus defined as the radiant power density incident upon a collecting surface (hence the subscript c). Radiant intensity, on the other hand, is the power per unit solid angle radiated from a source (particularly a point source or a source that has negligible area compared with the square of the viewing distance)^{7,8}:

$$\text{Radiant Intensity} \equiv I = \frac{\partial P}{\partial \omega_c}, \quad (\text{watts/steradian}). \quad (9)$$

And radiance, the radiometric analog to the more familiar photometric term brightness, is defined as the radiant power per unit solid angle per unit projected source area. The quantity radiance is used to characterize an extended source, that is, one that has an appreciable area compared with the square of the viewing distance.^{7,8} In differential form, we have

$$\text{Radiance} \equiv L = \frac{\partial^2 P}{\partial \omega_c \partial A_s \cos \theta_s}, \quad (\text{watts/steradian projected area}). \quad (10)$$

The radiance of a source is, in general, a function of position on the source and a function of the two angular variables θ and ϕ in conventional spherical coordinates.

Rearranging Eq. (10), which defines the radiance,

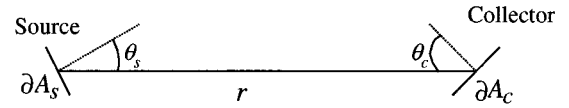


Fig. 2. Geometrical configuration used to demonstrate the fundamental theory of radiometry from which the quantity radiance is obtained.

and noting that $\partial \omega_c = \partial A_c \cos \theta_c / r^2$ and $\partial \omega_s = \partial A_s \cos \theta_s / r^2$, we obtain the double differential

$$\begin{aligned} \partial^2 P &= L \partial A_s \cos \theta_s \partial \omega_c \\ &= \frac{L \partial A_s \cos \theta_s \partial A_c \cos \theta_c}{r^2} \\ &= L \partial \omega_s \partial A_c \cos \theta_c, \end{aligned} \quad (11)$$

where r is the distance between the source and the collector, as illustrated in Fig. 2. Equation (11) can be considered to be the fundamental theorem of radiometry, as it describes the radiant-power transfer between an elemental source and an elemental collector.

If we consider the diffracting aperture of Fig. 1 as the source and the observation hemisphere as the collector, we can integrate Eq. (11) once with respect to the projected source area to obtain the diffracted intensity:

$$I = \frac{\partial P}{\partial \omega_c} = \int_{A_s} L \cos \theta_s \partial A_s. \quad (12)$$

Or, if we integrate the double differential of Eq. (11) once with respect to the source's solid angle $\partial \omega_s$, we see from Eq. (12) that the irradiance on the observation hemisphere ($\theta_c = 0$) is just the intensity divided by the square of the radius of the hemisphere:

$$\begin{aligned} E_c &= \frac{\partial P}{\partial A_c} = \int_{\omega_s} L(\theta, \phi, x, y) \partial \omega_s \\ &= \frac{1}{r^2} \int_{A_s} L(\theta, \phi, x, y) \cos \theta_s = \frac{1}{r^2} I(\theta, \phi). \end{aligned} \quad (13)$$

The direction cosines α , β , and γ are related to the angular variables θ and ϕ in conventional spherical coordinates by the following expressions:

$$\begin{aligned} \alpha &= \sin \theta \cos \phi, \\ \beta &= \sin \theta \sin \phi, \\ \gamma &= \cos \theta, \end{aligned} \quad (14)$$

and it can readily be shown that the differential solid angle $\partial \omega_c$ is given by

$$\partial \omega_c = \sin \theta \, d\theta \, d\phi = d\alpha \, d\beta / \gamma. \quad (15)$$

From Eq. (9) the total radiant power diffracted into the complete hemisphere can be expressed as

$$\begin{aligned} P_T &= \int_{\omega=0}^{2\pi} I(\theta, \phi) d\omega_c \\ &= \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} I(\theta, \phi) \sin \theta d\theta d\phi \\ &= \int_{\alpha=-1}^1 \int_{\beta=-(1-\alpha^2)^{1/2}}^{(1-\alpha^2)^{1/2}} I(\alpha, \beta) \frac{d\alpha d\beta}{\gamma}. \end{aligned} \quad (16)$$

Note that the entire hemisphere is represented by a unit circle in direction cosine space, and recall that only those plane-wave components that lie inside the unit circle are real and propagate. Those that lie outside of the unit circle are imaginary and are referred to as evanescent waves (and thus do not propagate).

The Rayleigh (Parseval) theorem from Fourier transform theory states that the integral over all space of the squared modulus of any function is equal to the integral over all space of the squared modulus of the Fourier transform of that function.^{1,9} We can therefore write

$$\begin{aligned} P_T &= \lambda^2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |U_0(\hat{x}, \hat{y})|^2 d\hat{x} d\hat{y} \\ &= \lambda^2 \int_{\alpha=-1}^1 \int_{\beta=-(1-\alpha^2)^{1/2}}^{(1-\alpha^2)^{1/2}} |\mathcal{F}\{U_0(\hat{x}, \hat{y})\}|^2 d\alpha d\beta. \end{aligned} \quad (17)$$

Because the right-hand sides of Eqs. (16) and (17) are equal, their integrands are equal:

$$I(\alpha, \beta) = \gamma \lambda^2 |\mathcal{F}\{U_0(\hat{x}, \hat{y})\}|^2. \quad (18)$$

And finally, from Eq. (12), we obtain the following general relation:

$$\int_{A_s} L(\alpha, \beta, x, y) \cos \theta_s dA_s = \gamma \lambda^2 |\mathcal{F}\{U_0(\hat{x}, \hat{y}; 0)\}|^2. \quad (19)$$

Now if the source is a uniformly illuminated diffracting aperture, as illustrated in Fig. 1, there is no dependence on position in the aperture:

$$\begin{aligned} I(\alpha, \beta) &= \int_{A_s} L(\alpha, \beta) \cos \theta_s dA_s \\ &= L(\alpha, \beta) \int_{A_s} \cos \theta_s dA_s \\ &\approx L(\alpha, \beta) A_s \cos \theta_s. \end{aligned} \quad (20)$$

Substituting expression (20) into Eq. (19), dividing by the area of the diffracting aperture A_s , and recalling

that $\gamma = \cos \theta_s$, we obtain the following expression for the diffracted radiance:

$$L(\alpha, \beta) = \frac{\lambda^2}{A_s} |\mathcal{F}\{U_0(\hat{x}, \hat{y}; 0)\}|^2. \quad (21)$$

We have thus shown that the squared modulus of the Fourier transform of the complex amplitude distribution emerging from the diffracting aperture yields the diffracted radiance and *not* the irradiance or the intensity. Furthermore, Eq. (21), unlike the more familiar Eq. (1), is not restricted to small diffraction angles. From the shift theorem of Fourier transform theory¹ it is clear that changes in the angle of incidence of the radiation illuminating the diffracting aperture will merely result in a shift of the radiance function in direction cosine space and an attenuation by the factor $\gamma_0 = \cos \theta_0$:

$$L(\alpha, \beta - \beta_0) = \gamma_0 \frac{\lambda^2}{A_s} |\mathcal{F}\{U_0(\hat{x}, \hat{y}; 0) \exp(i2\pi\beta_0\hat{y})\}|^2. \quad (22)$$

However, because the functional form does not change, the diffracted radiance is shift invariant in direction cosine space, and the simple Fourier techniques that have proven to be so useful in paraxial applications can be used for nonparaxial applications as well.

4. Renormalization to Conserve Energy in the Presence of Evanescent Waves

In extreme cases the above radiance-distribution function [Eq. (22)] will extend beyond the unit circle in direction cosine space. This can occur as a result of high-spatial-frequency content in the diffracting aperture that results in large diffracted angles or as a result of large incident angles that shift a radiance-distribution function of modest width such that it extends beyond the unit circle. In either case, evanescent waves are produced, and Eqs. (21) and (22) for radiance must be renormalized to conserve energy. As we see below, this is not done in a heuristic manner but is a direct result of the scalar diffraction theory.

Consider the example of a rotationally symmetric, on-axis, Gaussian radiance-distribution function, as illustrated in Fig. 3(a):

$$\begin{aligned} L(\alpha, \beta) &= \exp[-(\rho/\Delta\rho)^2], \\ \rho &= (\alpha^2 + \beta^2)^{1/2}, \\ \Delta\rho &= 0.2. \end{aligned} \quad (23)$$

We obtain the corresponding conventional diffracted intensity by applying expression (20) and noting that $\gamma = \cos \theta_s$ and plotting as a function of the diffraction angle, as illustrated in Fig. 3(b):

$$I(\theta) = \gamma A_s L(\alpha, \beta), \quad \theta = \sin^{-1}(\rho) = (1 - \rho^2)^{1/2}. \quad (24)$$

Clearly, both of the functions in Eqs. (23) and (24) are centered on the optical axis and contained entirely within the unit circle.

Suppose we now introduce a rather large 64° incident angle ($\beta_0 = 0.9$). The radiance-distribution function $L(\alpha, \beta)$ is shifted and attenuated by the constant γ_0 (without a change in the functional form) until it is centered on $\beta = -0.9$, as illustrated in Fig. 3(c). Note that a significant portion of this shifted radiance-distribution function $L(\alpha, \beta - \beta_0)$ extends beyond the unit circle in direction cosine space. Only that portion that lies inside the unit circle is real and contains radiant power or energy. Those plane-wave components of the diffracted wave field that lie outside of the unit circle are imaginary and are referred to as evanescent waves. If we now rewrite the Rayleigh theorem [previously expressed in Eq. (17)] for this case of non-normal incidence, we obtain

$$\begin{aligned} & \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |U_0(\hat{x}, \hat{y}) \exp(i2\pi\beta_0\hat{y})|^2 d\hat{x} d\hat{y} \\ &= \int_{\alpha=-1}^1 \int_{\beta=-(1-\alpha^2)^{1/2}}^{(1-\alpha^2)^{1/2}} |\mathcal{F}\{U_0(\hat{x}, \hat{y}) \exp(i2\pi\beta_0\hat{y})\}|^2 d\alpha d\beta. \end{aligned} \quad (25)$$

But

$$|U_0(\hat{x}, \hat{y}) \exp(i2\pi\beta_0\hat{y})|^2 = |U_0(\hat{x}, \hat{y})|^2; \quad (26)$$

hence with the help of Eqs. (17) and (21) the left-hand side of Eq. (25) can be written as

$$\begin{aligned} \frac{P_T}{\lambda^2} &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |U_0(\hat{x}, \hat{y})|^2 d\hat{x} d\hat{y} \\ &= \frac{A_s}{\lambda^2} \int_{\alpha=-1}^1 \int_{\beta=-(1-\alpha^2)^{1/2}}^{(1-\alpha^2)^{1/2}} L(\alpha, \beta) d\alpha d\beta. \end{aligned} \quad (27)$$

Similarly, with the help of Eq. (22) the right-hand side of Eq. (25) can be written as

$$\begin{aligned} \frac{P_T}{\lambda^2} &= \int_{-1}^1 \int_{(1-\alpha^2)^{1/2}}^{(1-\alpha^2)^{1/2}} |\mathcal{F}\{U_0(\hat{x}, \hat{y}; 0) \exp(i2\pi\beta_0\hat{y})\}|^2 d\alpha d\beta \\ &= \frac{1}{\gamma_0} \frac{A_s}{\lambda^2} \int_{-1}^1 \int_{-(1-\alpha^2)^{1/2}}^{(1-\alpha^2)^{1/2}} L'(\alpha, \beta - \beta_0) d\alpha d\beta. \end{aligned} \quad (28)$$

Now, by equating the right-hand sides of Eqs. (27) and (28), canceling λ^2 , and dividing by A_s , we obtain

$$\begin{aligned} \frac{P_T}{A_s} &= \int_{\alpha=-1}^1 \int_{\beta=-(1-\alpha^2)^{1/2}}^{(1-\alpha^2)^{1/2}} L(\alpha, \beta) d\alpha d\beta \\ &= \frac{1}{\gamma_0} \int_{\alpha=-1}^1 \int_{\beta=-(1-\alpha^2)^{1/2}}^{(1-\alpha^2)^{1/2}} L'(\alpha, \beta - \beta_0) d\alpha d\beta, \end{aligned} \quad (29)$$

where

$$L'(\alpha, \beta - \beta_0) = KL(\alpha, \beta - \beta_0) \quad (30)$$

is a normalized version of the original shifted radiance-distribution function $L(\alpha, \beta - \beta_0)$, as illustrated in Fig. 3(c). This normalization constant is

given by the ratio of the integral of the radiance-distribution function over infinite limits to the integral of the radiance-distribution function over the unit circle in direction cosine space and is equal to 1.36 for this particular example:

$$K = \frac{\int_{\alpha=-\infty}^{\infty} \int_{\beta=-\infty}^{\infty} L(\alpha, \beta - \beta_0) d\alpha d\beta}{\int_{\alpha=-1}^1 \int_{\beta=-(1-\alpha^2)^{1/2}}^{(1-\alpha^2)^{1/2}} L(\alpha, \beta - \beta_0) d\alpha d\beta} = 1.36. \quad (31)$$

Thus for those cases in which evanescent waves exist the diffracted radiance expressed in Eq. (22) must be normalized by the constant K so that energy will be conserved:

$$L'(\alpha, \beta - \beta_0) = \begin{cases} K\gamma_0 \frac{\lambda^2}{A_s} |\mathcal{F}\{U_0(\hat{x}, \hat{y}; 0) \exp(i2\pi\beta_0\hat{y})\}|^2, & \alpha^2 + \beta^2 \leq 1 \\ 0 & \alpha^2 + \beta^2 > 1 \end{cases}. \quad (32)$$

This normalization constant K differs from unity only if the radiance-distribution function extends beyond the unit circle in direction cosine space (i.e., only if evanescent waves are produced). The well-known Wood's anomalies that occur in diffraction-grating efficiency measurements are entirely consistent with this predicted renormalization in the presence of evanescent waves.^{10,11}

To calculate the radiant intensity to compare with experimental measurements (radiance is not a directly measurable quantity), one needs only to apply the Lambert cosine law (i.e., to multiply the radiance by $\gamma = \cos \theta_s$) and integrate over the source area, obtaining

$$I(\alpha, \beta) = \int_{A_s} L(\alpha, \beta) \cos \theta_s dA_s. \quad (33)$$

Again, if the source is a uniformly illuminated diffracting aperture, we obtain the diffracted intensity by multiplying Eq. (32) by $A_s \cos \theta_s$:

$$I(\alpha, \beta - \beta_0) = \begin{cases} K\gamma_0\gamma\lambda^2 |\mathcal{F}\{U_0(\hat{x}, \hat{y}; 0) \exp(i2\pi\beta_0\hat{y})\}|^2, & \alpha^2 + \beta^2 \leq 1 \\ 0 & \alpha^2 + \beta^2 > 1 \end{cases}. \quad (34)$$

Although the diffracted radiance can exhibit a discontinuity at the edge of the unit circle as noted above (for a Lambertian emitter the radiance is constant and drops discontinuously to zero at the edge of the unit circle), the cosine factor in Eq. (34) ensures that the diffracted intensity never exhibits such discontinuities. Figure 3(d) illustrates the diffracted-intensity profile of the current example in direction

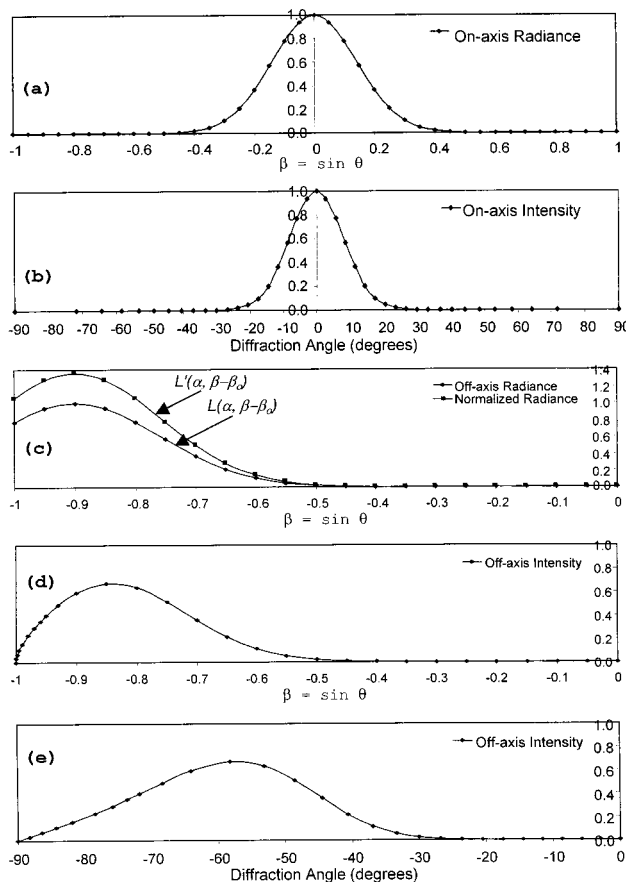


Fig. 3. (a) Profile of an on-axis Gaussian radiance distribution in direction cosine space. (b) The profile of the radiance distribution as a function of the diffraction angle that corresponds to (a). (c) Off-axis (normalized) radiance distribution resulting from a 64° incident angle. (d) The off-axis intensity distribution in direction cosine space that corresponds to (c). (e) Off-axis intensity distribution as a function of the diffraction angle.

cosine space. Note the asymmetry in this intensity profile that is characteristic of diffraction patterns at large incident angles. Figure 3(e) illustrates a profile of the same diffracted-intensity distribution when plotted as a function of the diffraction angle θ .

Diffraction gratings are inherently wide-angle devices that frequently are used with large oblique incident angles, and scattering effects from rough surfaces are another diffraction phenomenon that is not limited to the paraxial region. The two practical examples in Sections 5 and 6 show that many of the limitations usually attributed to scalar diffraction theory are really the result of the unnecessary paraxial approximation and not a limitation inherent to scalar theory.

5. Example 1: Sinusoidal Phase Gratings

Scalar diffraction theory frequently is considered inadequate for predicting diffraction efficiencies for grating applications for which $\lambda/d > 0.1$. It has also been stated that scalar theory imposes energy on the evanescent diffracted orders.¹² These notions, as

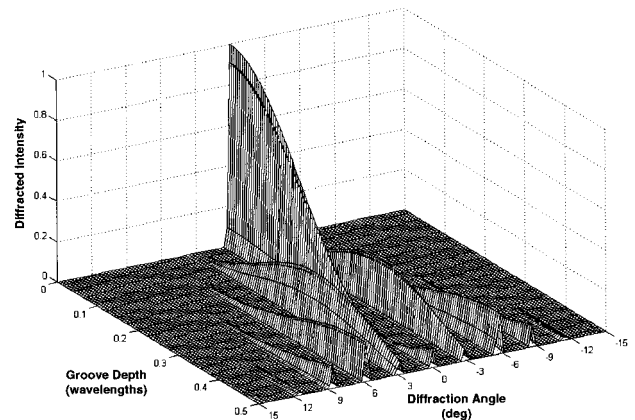


Fig. 4. Diffracted-intensity distribution as predicted by the paraxial model of Eq. (35) for a sinusoidal reflection grating with a period of $d = 20\lambda$ and operating at normal incidence.

well as several other common misconceptions, are driven more by an unnecessary paraxial approximation than by the scalar limitation.

A. Paraxial Model of Sinusoidal Phase Gratings

Following Goodman,² we obtain the paraxial equation for the diffracted intensity from a sinusoidal phase grating illuminated with a narrow Gaussian beam of width b (the radius at which the field drops to $e^{-\pi}$ of its peak value):

$$I(\theta_x, \theta_y) = I_0 \sum_{m=-\infty}^{\infty} J_m^2(a/2) \text{Gaus}^2 \left[\frac{b}{\lambda} (\theta_x - m\lambda/d, \theta_y) \right]. \quad (35)$$

Here we use Gaskill's symbolic notation for the Gaussian function¹:

$$\text{Gaus} \left(\frac{b\theta}{\lambda} \right) = \exp \left[-\pi \left(\frac{b\theta}{\lambda} \right)^2 \right]. \quad (36)$$

Figure 4 illustrates the resultant diffracted-intensity distribution as a function of the diffraction angle θ_x and the groove depth h for a sinusoidal reflection grating ($a = 4\pi h/\lambda$) with a period of $d = 20\lambda$ operating at normal incidence.

The maximum value of $J_1^2(a/2)$ is 0.3386 and occurs for $a = 3.68$ ($h = 0.293\lambda$). The diffraction efficiency of the first few orders for this value of a are listed in Table 1. Note that the energy falls off rap-

Table 1. Diffraction Efficiencies for a Coarse (Paraxial Regime) Sinusoidal Phase Grating Optimized for Maximum Efficiency in the ± 1 Orders

Diffracted Orders	Diffraction Efficiencies
0	1.003×10^{-1}
± 1	3.386×10^{-1}
± 2	9.970×10^{-2}
± 3	1.093×10^{-2}
± 4	6.320×10^{-4}

idly, with 99.88% of the diffracted radiation contained in diffracted orders $|m| \leq 3$. This paraxial model is accurate for only very coarse gratings ($d \gg \lambda$) and leads to the common misconception that it is impossible to get more than 33.86% of the incident energy into the first diffracted order with a sinusoidal phase grating.

If we use Eq. (35) to model the behavior of fine (small-spatial-period) gratings whose diffracted orders do not fall within the paraxial region, we obtain incorrect results that lead to a variety of other misconceptions. For example, a sinusoidal reflection grating with a period of $d = 1.3\lambda$ results in the predicted intensity distribution illustrated in Fig. 5.

There are several features of this predicted intensity distribution that are inaccurate: (i) The small-angle approximation to the grating equation inherent in Eq. (35) leads to the erroneous prediction of equally spaced diffracted orders. (ii) When a diffracted order of finite width is located near $\pm 90^\circ$ the predicted diffracted intensity (or irradiance) exhibits a discontinuity that is an unacceptable nonphysical phenomenon. (iii) The predicted angular width of the diffracted orders does not indicate the broadening that occurs with increasing diffraction angle.¹¹ (iv) The diffraction efficiencies of the propagating orders are not adjusted to ac-

Hence a sinusoidal reflection grating with a period of $d = 1.95\lambda$ results in the predicted radiance distribution illustrated in Fig. 6. This distribution is very similar to that shown in Fig. 5 except that here we plotted the diffracted radiance in direction cosine space. Note that the diffracted orders are equally spaced (and have equal widths) in direction cosine space.¹³ Also, discontinuities in the radiance are allowed at $\alpha = \pm 1$ ($\theta = \pm 90^\circ$). Recall that, for a Lambertian reflector, the radiance is constant over the entire hemisphere with a discontinuity at $\pm 90^\circ$. Also, this diffracted-radiance distribution has been renormalized in accordance with Eq. (31). This renormalization ensures that all the transmitted energy is contained in the propagating plane-wave components.

To calculate the diffracted-intensity distribution, we merely apply the Lambert cosine law (i.e., we multiply the radiance by $\gamma = \cos \theta$) and integrate over the source area to obtain

$$I(\alpha, \beta) = \int_{A_s} L'(\alpha, \beta) \text{Gaus}^2(r/b) \cos \theta \, dA_s \\ = L'(\alpha, \beta) \cos \theta \int_{A_s} \text{Gaus}^2(r/b) dA_s. \quad (38)$$

The integral in Eq. (38) is equal to $b^2/2$ hence

$$I(\alpha, \beta) = \begin{cases} K\gamma E_0 \hat{b}^4 \lambda^2 \sum_{m=-\infty}^{\infty} J_m^2(a/2) \text{Gaus}^2[\hat{b}(\alpha - m/\hat{d}, \beta)] & \alpha^2 + \beta^2 \leq 1 \\ 0 & \alpha^2 + \beta^2 > 1 \end{cases}, \quad (39)$$

count for the Wood's anomaly effect of redistributing the energy associated with evanescent orders among the remaining propagating orders.¹⁰ This has led some authors to state that scalar theory imposes energy on the evanescent diffracted orders.¹² We show that it is not scalar theory but the unnecessary paraxial approximation that leads to these misconceptions.

B. Wide-Angle (Nonparaxial) Behavior of Sinusoidal Phase Gratings

Applying the results of our new nonparaxial scalar diffraction theory to the case of a Gaussian beam of width b (the radius at which the field drops to $e^{-\pi}$ of its peak value) normally incident upon a sinusoidal phase grating with a period of $d \ll b$ causes Eq. (32) to become

$$L'(\alpha, \beta) = \begin{cases} K2E_0 \hat{b}^2 \sum_{m=-\infty}^{\infty} J_m^2(a/2) \text{Gaus}^2[\hat{b}(\alpha - m/\hat{d}, \beta)] & \alpha^2 + \beta^2 \leq 1 \\ 0 & \alpha^2 + \beta^2 > 1 \end{cases}, \quad (37)$$

Equation (39) causes the intensity distribution to fall gracefully to zero at $\alpha = \pm 1$ (or $\theta = \pm 90^\circ$). The predicted intensity distribution as a function of the groove depth and the diffracted angle is illustrated in Fig. 7.

Note that, when plotted as a function of the diffracted angle, the diffracted orders are no longer equally spaced, and they become progressively wider with an increasing diffraction angle. In fact, for large diffraction angles the peak (or centroid) of the diffracted order is shifted from the position that is predicted by the grating equation.¹¹

6. Example 2: Surface-Scatter Phenomena

The effects of light scattered from optical surfaces have received considerable attention in recent years. In many applications it is not only the amount of scattered light but also the direction of the scattered radi-

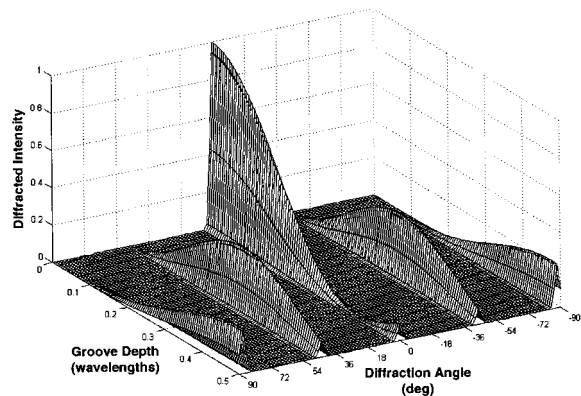


Fig. 5. Diffracted-intensity distribution as predicted by the paraxial model of Eq. (35) for a sinusoidal reflection grating with a period of $d = 1.3\lambda$, which is an invalid operating regime.

ation that play a crucial role in the performance of the optical system. This is particularly true for two distinct types of applications: (i) the design and the analysis of stray-light-rejection systems required by optical instruments that are used to view a relatively faint target in the vicinity of a much brighter object, and (ii) the fabrication of supersmooth surfaces for high-resolution x-ray and extreme-UV imaging systems. The optical performance in the first application is degraded by wide-angle scatter caused by the surface microroughness (or surface finish), and the optical performance in the second application is degraded predominantly by the small-angle scatter caused by longer-range mid-spatial-frequency surface irregularities.^{14–16}

The Rayleigh–Rice vector-perturbation theory agrees well with experimental wide-angle scatter measurements from smooth surfaces for arbitrary incident angles.^{17,18} However, not all applications of interest satisfy the smooth-surface approximation. The Beckmann–Kirchhoff (B-K) scalar diffraction theory of surface-scatter phenomena is valid for rougher

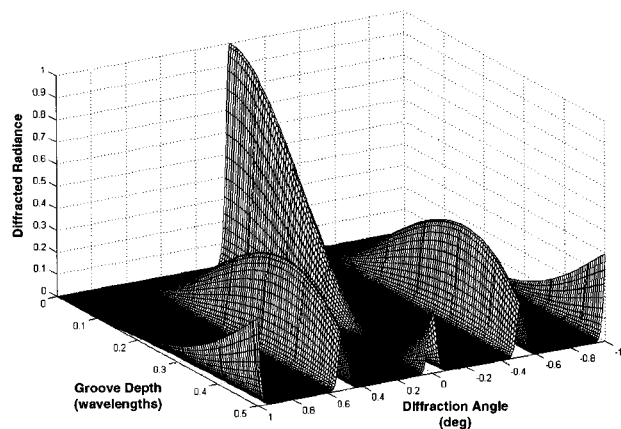


Fig. 6. Plot of the diffracted-radiance distribution (in direction cosine space) as a function of the groove depth. The distribution was produced by a sinusoidal reflection grating with a period of $d = 1.95\lambda$.

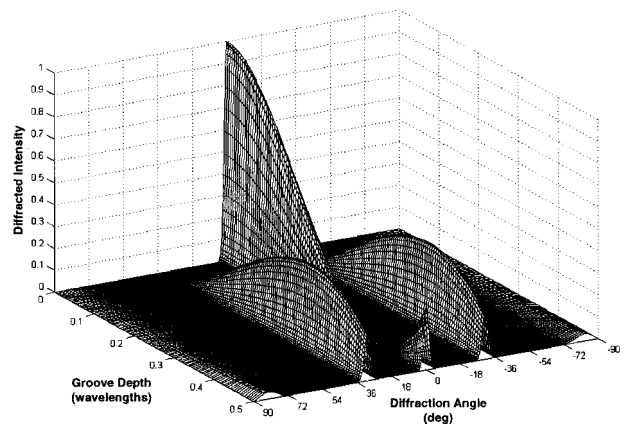


Fig. 7. Plot of the diffracted-intensity distribution as a function of the groove depth and the diffraction angle. The distribution was produced by a sinusoidal reflection grating with a period of $d = 1.95\lambda$.

surfaces but contains a paraxial (small-angle) assumption that limits its ability to handle wide-angle scattering and large angles of incidence accurately.^{18,19}

A linear systems approach to modeling surface-scatter phenomena was suggested by Shack and developed by Harvey in the 1970's. Based on the empirical observation that wide-angle scattering behavior is shift invariant in direction cosine space, this treatment resulted in a surface transfer function (STF) that relates scattering behavior to surface topography.^{20,21} This STF led to a modest following among the radiometric community of bidirectional-reflectance-distribution function curves plotted in the Harvey–Shack $\beta - \beta_0$ format.^{22–24} During the 1980's this STF was generalized to include (i) the effects of small-angle scatter caused by mid-spatial-frequency surface irregularities that span the gap between the traditional figure and finish errors, and (ii) the extremely large incident angles inherent to grazing-incidence Wolter type I x-ray telescopes.¹⁵

The nonparaxial scalar diffraction theory presented in Sections 3 and 4 of this paper combined with this linear systems treatment of surface-scatter phenomena are used below to explain some nonintuitive surface-scatter effects that are not readily explained by the classical B-K scattering theory.

A. Nonintuitive Experimental Scattering Data

A detailed experimental investigation of light scattering from characterized random surfaces was reported by O'Donnell and Mendez²⁵ in 1987. Although the paper dealt primarily with enhanced backscatter phenomena that were due to multiple scattering, preliminary scattering measurements were made on a wide-scale, well-characterized, highly conducting surface with a Gaussian surface autocovariance (ACV) function

$$C\left(\frac{x}{l}, \frac{y}{l}\right) = \sigma_s^2 \exp\left[-\left(\frac{r}{l}\right)^2\right]. \quad (40)$$

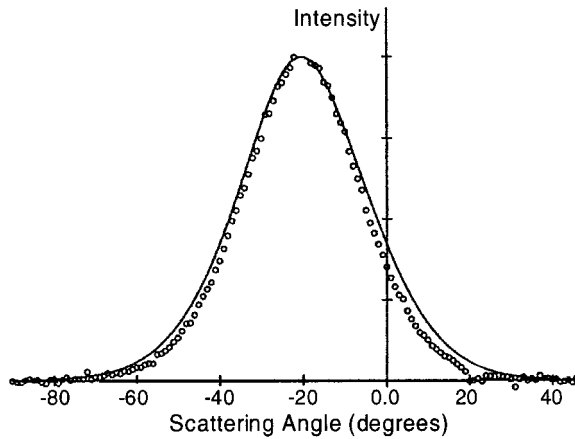


Fig. 8. Experimental data for modest incident angles compared with the predictions from the B-K theory with $\lambda = 0.6328$, $10.6 \mu\text{m}$ and $\theta_i = 20^\circ$. The experimental data always exhibited a slightly narrower distribution. Reprinted from Ref. 25 with the permission of the authors.

The measured rms surface roughness was $\sigma_s = 2.27 \mu\text{m}$, and the measured ACV length (the e^{-1} half-width of the ACV function) was $l = 20.9 \mu\text{m}$. For the small slopes represented by these parameters, multiple-scattering effects are negligible, and conventional high-angular-resolution scattering data were measured for two wavelengths, $\lambda = 0.6328$, $10.6 \text{ \AA}$, and two angles of incidence, $\theta_i = 20^\circ$, 70° . The experimental measurements were compared with classical B-K theory for the mean scattered intensity for a perfectly conducting Gaussian surface.¹⁹

Several puzzling effects were observed when comparing the experimental scattered-intensity measurements with the classical B-K scattering theory. For modest angles of incidence the B-K solution agrees quite well with the experimental data; however, a persistent tendency for the data to be narrower than predicted by theory was observed (see Fig. 8). It was suggested by the authors that this minor departure might be due to a resonance effect because $l \sim \lambda$; however, this was not verified.

For angles of incidence greater than 50° substantial disagreement between the experimental measurements and the measured data were observed. Figure 9 illustrates the data for a wavelength of $\lambda = 0.6328 \mu\text{m}$ and an angle of incidence of $\theta_i = 70^\circ$. The experimental data are highly asymmetrical about the specular direction (note that there is no specular beam at this wavelength) and drop smoothly to zero at -90° . The B-K theory predicts a symmetrical intensity distribution about the specular direction, although it is nonphysical in that it exhibits a discontinuity at -90° . The authors suggest that the B-K theory fails at this angle of incidence primarily because of shadowing and multiple-scattering effects and state that they are unaware of any available theory that compares with their measured data. We show that it is a funda-

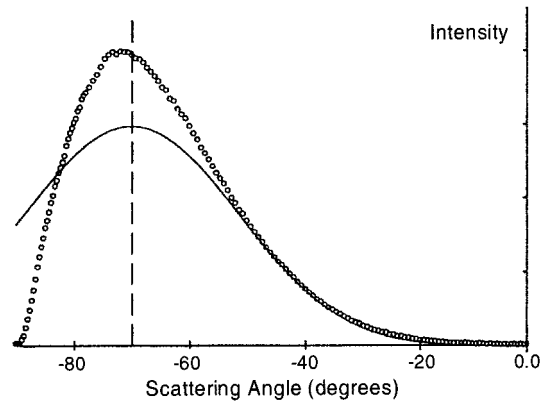


Fig. 9. Illustration of the drastic departure exhibited between the experimental data and the theoretical predictions for large incident angles with $\lambda = 0.6328 \mu\text{m}$ and $\theta_i = 70^\circ$. Reprinted from Ref. 25 with the permission of the authors.

mental misunderstanding of nonparaxial surface-scatter phenomena that causes this discrepancy and that a simple scalar diffraction theory of surface-scatter phenomena will indeed explain these results.

Another unusual effect is illustrated in Fig. 10. For a wavelength of $\lambda = 10.6 \mu\text{m}$ and an angle of incidence of $\theta_i = 70^\circ$ the measured scattering function does not peak in the specular direction. At this wavelength there is a weak specular component; however, the data points influenced by it have been omitted. The peak of the measured scattering function lies substantially inside of the specular beam. The B-K theory for this wavelength no longer exhibits symmetry about the specular direction and shows a similar shift of the peak of the diffuse component. Again the authors had no explanation for this shifting of the peak of the scattering function from the specular direction.

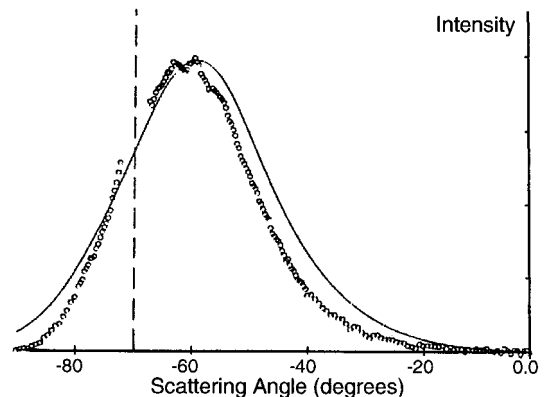


Fig. 10. Peak of the diffusely scattered component, which lies inside of the specular direction for both the experimental data and the B-K theory with $\lambda = 10.6 \mu\text{m}$ and $\theta_i = 70^\circ$. Reprinted from Ref. 25 with the permission of the authors.

B. Scattered-Radiance Distribution

The generalized STF for modeling scattering effects for large incident angles is given by¹⁵

$$H_s(\hat{x}, \hat{y}) = \exp \left\{ -(4\pi \cos \theta_0 \hat{\sigma}_s)^2 \right. \\ \left. \times \left[1 - C_s \left(\frac{\hat{x}}{\hat{l}}, \frac{\hat{y}}{\hat{l} \cos \theta_0} \right) / \sigma_s^2 \right] \right\}. \quad (41)$$

Equation (41) for the STF can be expressed in terms of the quantities A and B :

$$H_s(\hat{x}, \hat{y}) = A + BQ(\hat{x}, \hat{y}), \quad (42)$$

where

$$A = \exp[-(4\pi \cos \theta_0 \hat{\sigma}_s)^2], \quad (43)$$

$$B = 1 - \exp[-(4\pi \cos \theta_0 \hat{\sigma}_s)^2], \quad (44)$$

$$Q(\hat{x}, \hat{y}) = \frac{\exp \left[(4\pi \cos \theta_0 \hat{\sigma}_s)^2 C_s \left(\frac{\hat{x}}{\hat{l}}, \frac{\hat{y}}{\hat{l} \cos \theta_0} \right) / \sigma_s^2 \right] - 1}{[\exp(4\pi \cos \theta_0 \hat{\sigma}_s)^2 - 1]}. \quad (45)$$

The significance of the above interpretation of the equivalent transfer function of the scattering surface is shown by the inferred properties of the corresponding angle-spread function. Because the transfer function is the sum of two separate components, the angle-spread function of the scattering surface is the sum of the Fourier transforms of the two component functions:

$$\mathcal{F}(\alpha, \beta; \hat{r}) = \mathcal{F}\{H_s(\hat{x}, \hat{y})\} = A\delta(\alpha, \beta; \hat{r}) + S(\alpha, \beta; \hat{r}), \quad (46)$$

where the scattering function is given by

$$S(\alpha, \beta; \hat{r}) = B\mathcal{F}\{Q(\hat{x}, \hat{y})\}. \quad (47)$$

The constant component of the STF transforms into a delta function that represents a specular beam, and the other component transforms into a scattering function. Hence the scattering surface reflects an incident beam of light as a specularly reflected beam of diminished intensity surrounded by a halo of scattered light. Furthermore, from the central-ordinate theorem of Fourier transform theory, the angle-spread function is normalized to unit volume, and the relative power distributions between the specular component and the scattered component of the angle-spread function are given by the quantities A and B , respectively.

The angle-spread function defined in Eq. (46) is given by the Fourier transform of the STF that was defined as the normalized autocorrelation of the complex pupil function. From the autocorrelation theorem of Fourier transform theory the angle-spread function is also given by the squared modulus of the Fourier transform of the complex pupil function.^{1,9} From Eq. (21) in Section 3, we thus know that the

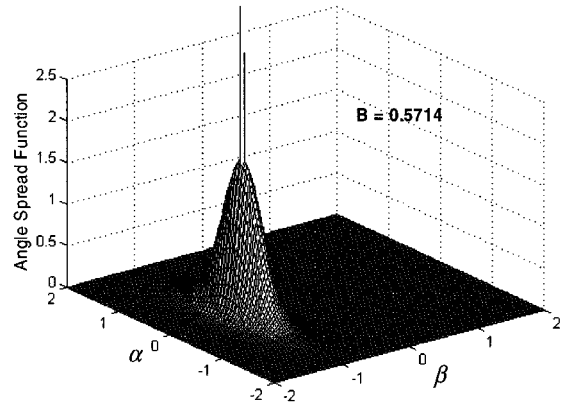


Fig. 11. Off-axis angle-spread function associated with the surface ACV function of Eq. (40) with $\lambda = 10.6 \mu\text{m}$, $\theta_0 = 70^\circ$, and $B = 0.5714$.

angle-spread function is a scattered-radiance-distribution function.

Although the scattering process is clearly not shift invariant with respect to the incident angle, the linear systems approach to modeling surface-scatter phenomena is still valid, provided we use a different STF for each incident angle, as indicated by Eq. (41). This practice is already well established by use of different modulation transfer functions for each field angle when characterizing imaging systems with field-dependent aberrations.

Figure 11 illustrates the off-axis angle-spread function associated with the surface ACV function of Eq. (40) for a beam with a $10.6\text{-}\mu\text{m}$ wavelength incident at an angle of 70° . Because the incident beam underfills the scattering surface, the illuminated footprint on the scattering surface is elongated in the plane of incidence, and the associated angle-spread function is correspondingly foreshortened, as indicated. For this large incident angle only 57% of the incident light is scattered, leaving 43% of the incident energy in the specular beam. Also, this off-axis angle-spread function extends beyond the unit circle

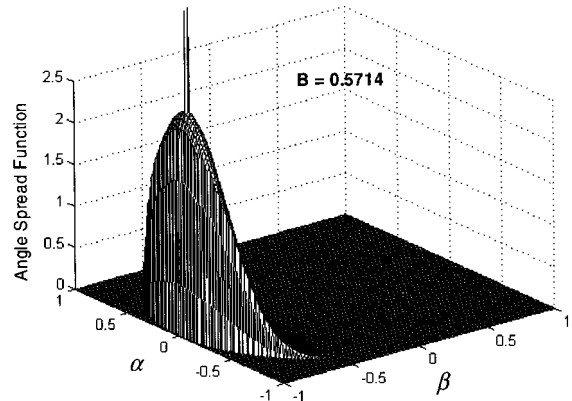


Fig. 12. Three-dimensional isometric plot of the truncated and the renormalized angle-spread function for an incident angle of 70° with $\lambda = 10.6 \mu\text{m}$, $B = 0.5714$, and $K = 1.4112$.

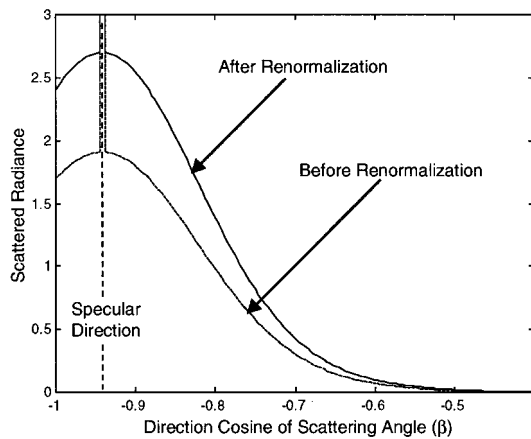


Fig. 13. Comparison of the scattered-radiance profiles in the plane of incidence before and after renormalization. $K = 1.4177$.

in direction cosine space. Evanescent waves are thus produced, and the resulting angle-spread function must be truncated and renormalized, as dictated by Eq. (30).

The renormalization constant, $K = 1.4112$, is obtained by calculation of the total volume of the radiance-distribution function illustrated in Fig. 11 and then by division by that portion of the volume that lies within the unit circle in direction cosine space. Figure 12 illustrates a three-dimensional isometric plot of this renormalized scattered-radiance-distribution function, and Fig. 13 compares profiles of the predicted angle-spread function before and after renormalization.

C. Scattered-Intensity Distribution

We obtain the scattered intensity (the radiant power per unit solid angle) by merely applying the Lambert cosine law, i.e., by multiplying by the cosine of the scattering angle, $\gamma = \cos \theta$, and integrating over the illuminated area on the scattering surface, as described by Eq. (33). This results in the scattered-intensity distribution illustrated in Fig. 14. The scattered-intensity distribution will thus always fall

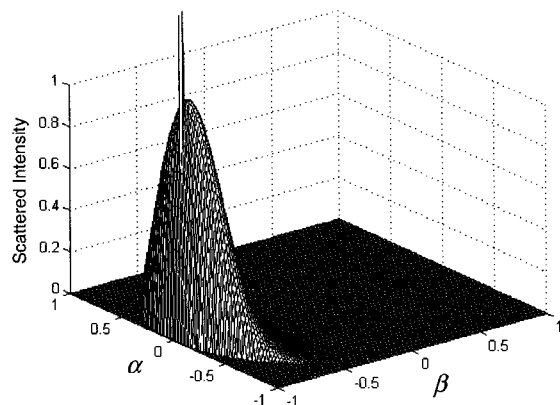


Fig. 14. Scattered intensity associated with the surface ACV function of Eq. (40) for an incident angle of 70° with $\lambda = 10.6 \mu\text{m}$.

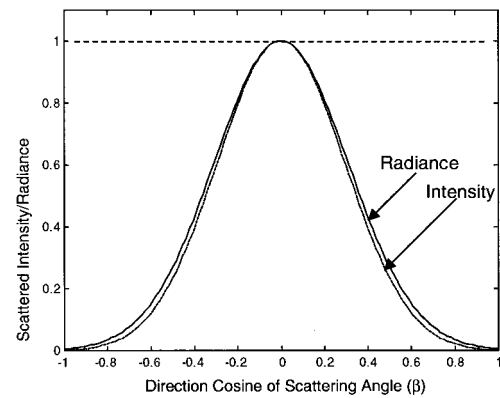


Fig. 15. Predicted curves of the scattered radiance and the scattered intensity for normal incidence, $\lambda = 0.6328 \mu\text{m}$, $\theta_0 = 0.0^\circ$, and $B = 0.00$.

gracefully to zero at the edge of the unit circle in direction cosine space. Also, applying the Lambert cosine law has the effect of shifting the peak of the intensity distribution away from the specular direction toward the surface normal.

We now begin to understand the nonintuitive figures of O'Donnell and Mendez presented in Subsection 6.A. The experimental data are indeed the scattered intensity, but the predictions from the B-K theory are a better representation of the scattered radiance. Beckmann, as presented in Ref. 19, does not explicitly state what radiometric quantity he is predicting; however, his expressions are always equated with the squared modulus of the scattered electric field, which is usually assumed to be proportional to the intensity (or the irradiance). Recall that we showed in Section 3 that the squared modulus of the Fourier transform of the complex amplitude distribution (which is proportional to the electric field) emerging from a diffracting aperture (or a scattering surface) yields the diffracted-scattered radiance (not the irradiance or the intensity). Furthermore, this was shown to be true for wide-angle (nonparaxial) diffraction at large incident angles. Because many diffraction-scattering theories have a paraxial limitation and there is no difference between radiance and intensity in the paraxial region ($\cos \theta \sim 1$ for small θ), it has not been widely recognized that radiance is the fundamental quantity that is predicted accurately by scalar diffraction theory. After we recognize that the theoretical predictions represent scattered radiance and the experimental measurements are indeed scattered intensity, the discrepancies indicated in Figs. 8–10 become not only understandable but intuitive.

For small angles of incidence applying the Lambert cosine law to the scattered-radiance distribution will always result in a slightly narrower scattered-intensity distribution. Figure 15 shows a comparison between the scattered radiance and the scattered intensity as predicted for a beam normally incident upon the scattering surface characterized by the ACV

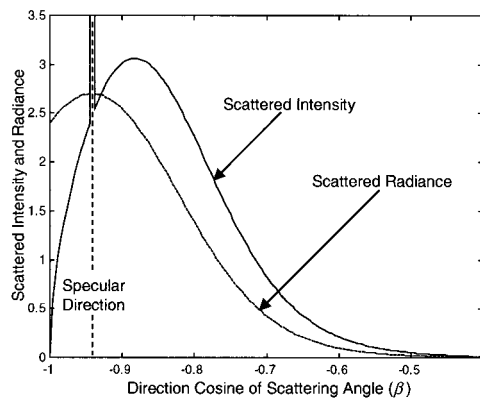


Fig. 16. Comparison of the predicted scattered radiance and the predicted scattered intensity for a large incident angle with $\lambda = 10.6 \mu\text{m}$ and $\theta_0 = 70^\circ$.

function of Eq. (40). Recall that this was reported by O'Donnell and Mendez²⁵ as a persistent effect that showed up repeatedly in their measurements.

For large angles of incidence the scattered-radiance function is substantially shifted and abruptly truncated by the unit circle in direction cosine space, as was shown in Fig. 13. The scattered radiance and the scattered intensity, as predicted by our nonparaxial scalar diffraction theory, are compared in Fig. 16 for a beam with a $10.6\text{-}\mu\text{m}$ wavelength reflected (with an incident angle of 70°) from the scattering surface characterized by the ACV function of Eq. (40). These functions were normalized to the same volume; it is thus a similar comparison to that presented by O'Donnell and Mendez²⁵ in Fig. 9. Note that the scattered-radiance distribution is symmetric about the specular beam, as noted by O'Donnell and Mendez concerning the B-K prediction.²⁵ The scattered-radiance distribution is also discontinuous at a scattering angle of -90° . Such behavior is allowed by radiance; however, it would be nonphysical behavior for the scattered intensity. The scattered intensity falls smoothly to zero at a scattering angle of -90° ; however, it is not symmetric about the specular direction, and, in fact, the peak of the scattered intensity is not at the same angle as that of the specular beam. All the nonintuitive features of wide-angle surface-scattering data for large incident angles are thus at least qualitatively explained by this linear systems treatment of surface-scatter phenomena as a simple scalar diffraction process.

D. Comparison of Theoretical Predictions with Experimental Data

The preceding discussion has hopefully provided insight and understanding concerning some previously unexplained nonintuitive behavior of wide-angle surface-scatter phenomena for large incident angles. However, we now show that this simple scalar diffraction theory of surface-scatter phenomena provides not only qualitative behavior and a basic

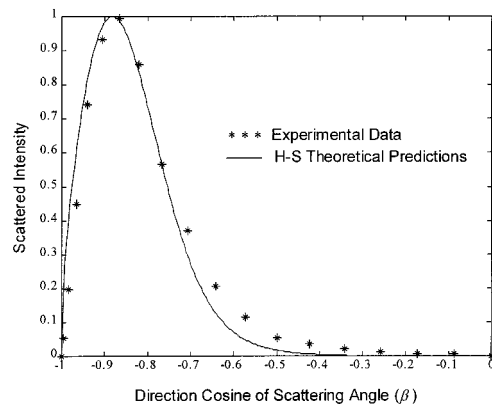


Fig. 17. Comparison of the Harvey-Shack theoretical predictions and the experimental data for large incident angles with $\lambda = 10.6 \mu\text{m}$ and $\theta_0 = 70^\circ$.

understanding that is difficult to obtain from more rigorous electromagnetic (vector) treatments but also provides remarkably accurate quantitative results. Figure 17 shows a direct comparison of predictions by our theory with the O'Donnell-Mendez experimental data for the case of $\lambda = 10.6 \mu\text{m}$ and $\theta_0 = 70^\circ$ for the scattering surface characterized by the ACV function described by Eq. (40).

We drew a smooth curve through the experimental scattering data from an enlarged copy of Fig. 10 and then sampled that curve at 5° intervals. Although the scattering data were obtained as angular data recorded with a precision goniometric instrument, we replotted the experimental data ($\beta = \sin \theta$) and made our comparison in direction cosine space because that is the natural space in which all diffraction, scattering, and propagation phenomena are most simply modeled.¹³ Finally, we normalized both the measured scattered-intensity data and the theoretically predicted scattered-intensity function to a peak value of unity.

The agreement is remarkably good, especially since this is a relatively rough surface ($\sigma = 2.27 \mu\text{m}$ and $\lambda = 10.6 \mu\text{m}$) illuminated at a very large angle of incidence. However, at this angle of incidence there is a specular beam that contains almost 43% of the total reflected radiant power ($B = 0.5714$). Although the Harvey-Shack theory is a scalar diffraction theory, there is no explicit smooth-surface approximation in the derivation of the STF.

Note that the major departure from the experimental data is at the knee of the curve. We achieved excellent agreement even in the knee of this curve with a modified B-K scattering theory, which was reported in a separate paper;²⁶ however, it does not provide the insight of the current treatment. We also achieved some improvement in the Harvey-Shack model by correcting for specific aberration terms (predominantly astigmatism) that were ignored in the derivation of Eq. (41).^{3,5} The effect of these aberrations (inherent to the diffraction process) on the scattered-intensity predictions is also a topic of

continuing research and is planned for another report.

7. Summary and Conclusions

Most textbook treatments of the propagation of diffracted wave fields include a paraxial (small-angle) assumption. Although this assumption may be reasonable for imaging systems with clear apertures, it is clearly inappropriate when dealing with diffraction gratings (inherently wide-angle devices) or scattering surfaces, both of which are frequently used with large oblique incident angles.

After a brief historical review of the linear systems formulation of scalar diffraction theory it has been shown that diffracted radiance, not irradiance or intensity, is the fundamental radiometric quantity that exhibits shift-invariant behavior in direction cosine space. We emphasized that, for nonparaxial applications in which the radiance-distribution function appears to extend beyond the unit circle in direction cosine space, only that portion that lies inside the unit circle is real and contains radiant power or energy. Hence the radiance function must be renormalized to conserve energy. Furthermore, this is not merely a heuristic correction; it is a direct result of the Rayleigh (or the Parseval) theorem in the mathematical development of a nonparaxial scalar diffraction theory. To convert from radiance to diffracted intensity, one then merely applies the Lambert cosine law.

Wide-angle diffraction from sinusoidal phase gratings has been discussed in some detail. The revelation that diffracted radiance is shift invariant in direction cosine space is particularly productive in terms of the insight and the understanding gained from this treatment. It clearly predicts Wood's anomalies and dispels the erroneous notion that scalar theory imposes energy on the evanescent orders.¹²

Scattering effects from rough surfaces are also diffraction phenomena that are not limited to the paraxial region. A discussion of this application not only provides physical insight and understanding of a variety of nonintuitive scattering effects reported previously in the literature for large incident and scattering angles but also provides experimental validation of the theory presented in Sections 3 and 4.

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